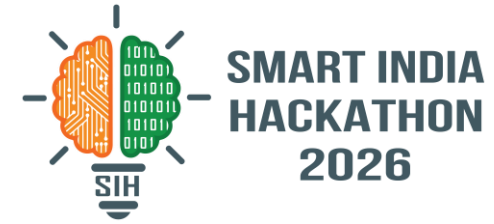
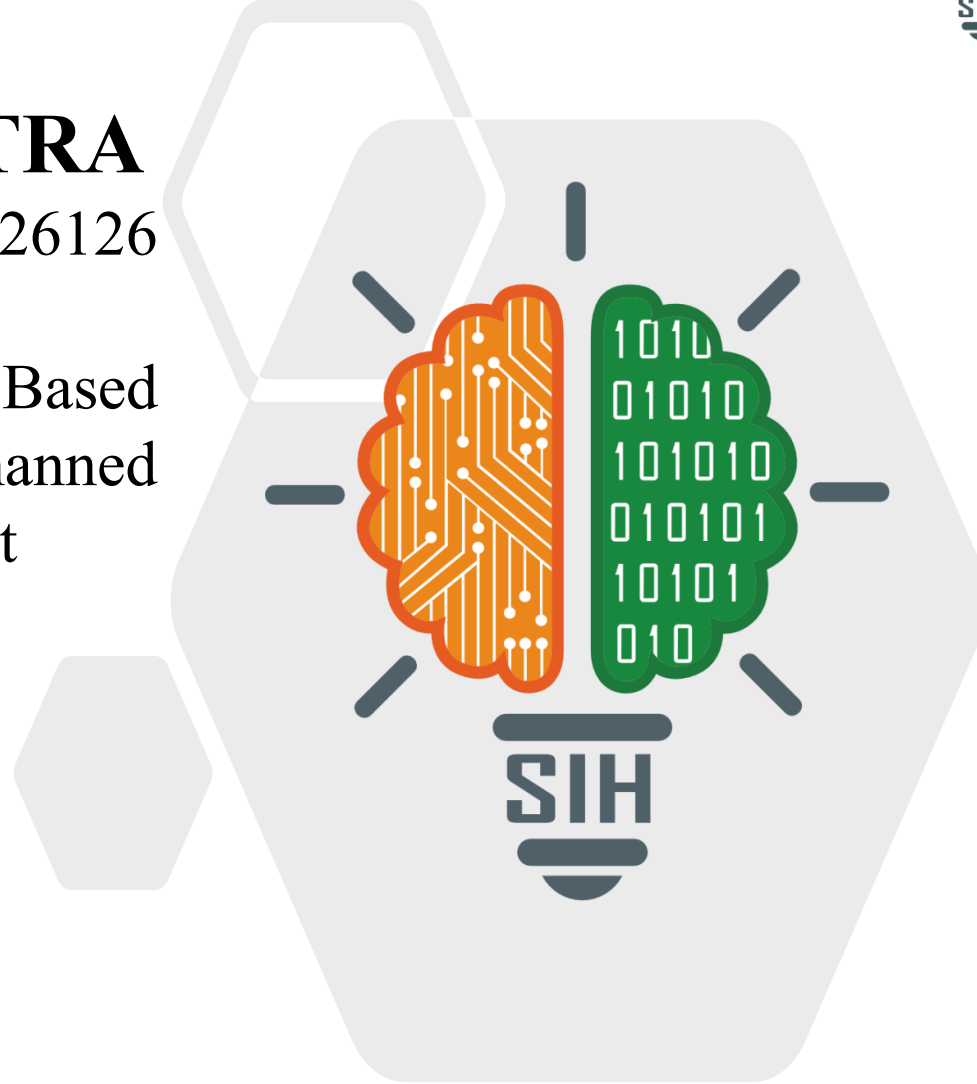


SMART INDIA HACKATHON 2026



VISTRA

- **Problem Statement ID** – 26126
- **Problem Statement Title-** Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor Environment
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 142936
- **Team Name -** NEXORAS

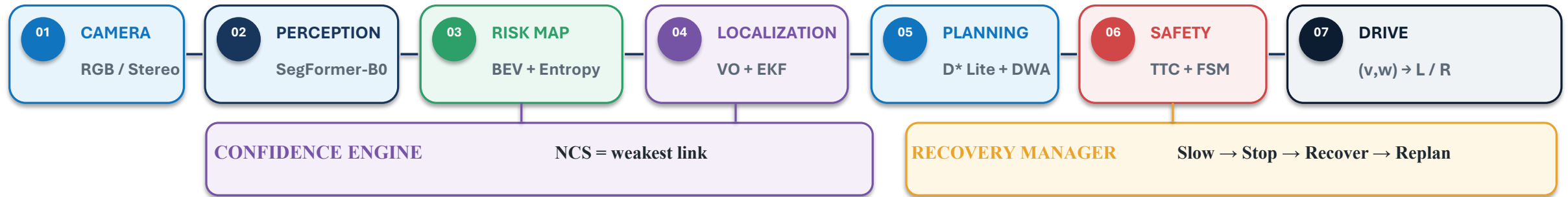


❖ PROPOSED SOLUTION

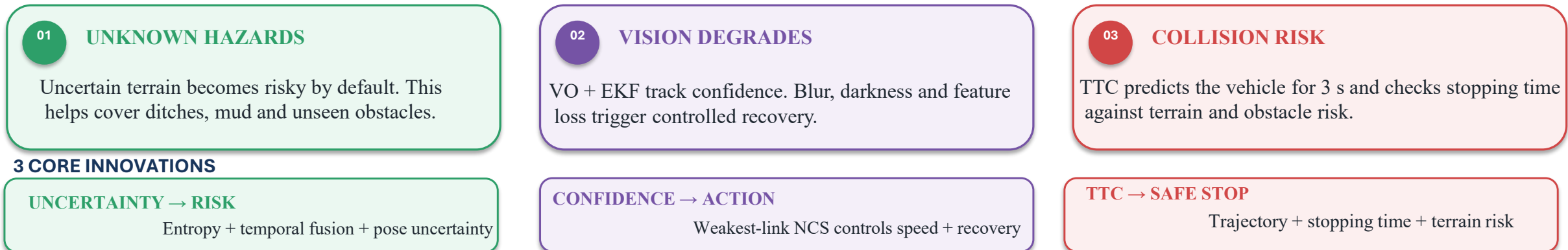
From pixels → confidence → safe driving

VISTRA carries uncertainty from the camera to the wheel commands — then slows, recovers, replans or stops.

HOW VISTRA THINKS



HOW IT ADDRESSES THE PROBLEM



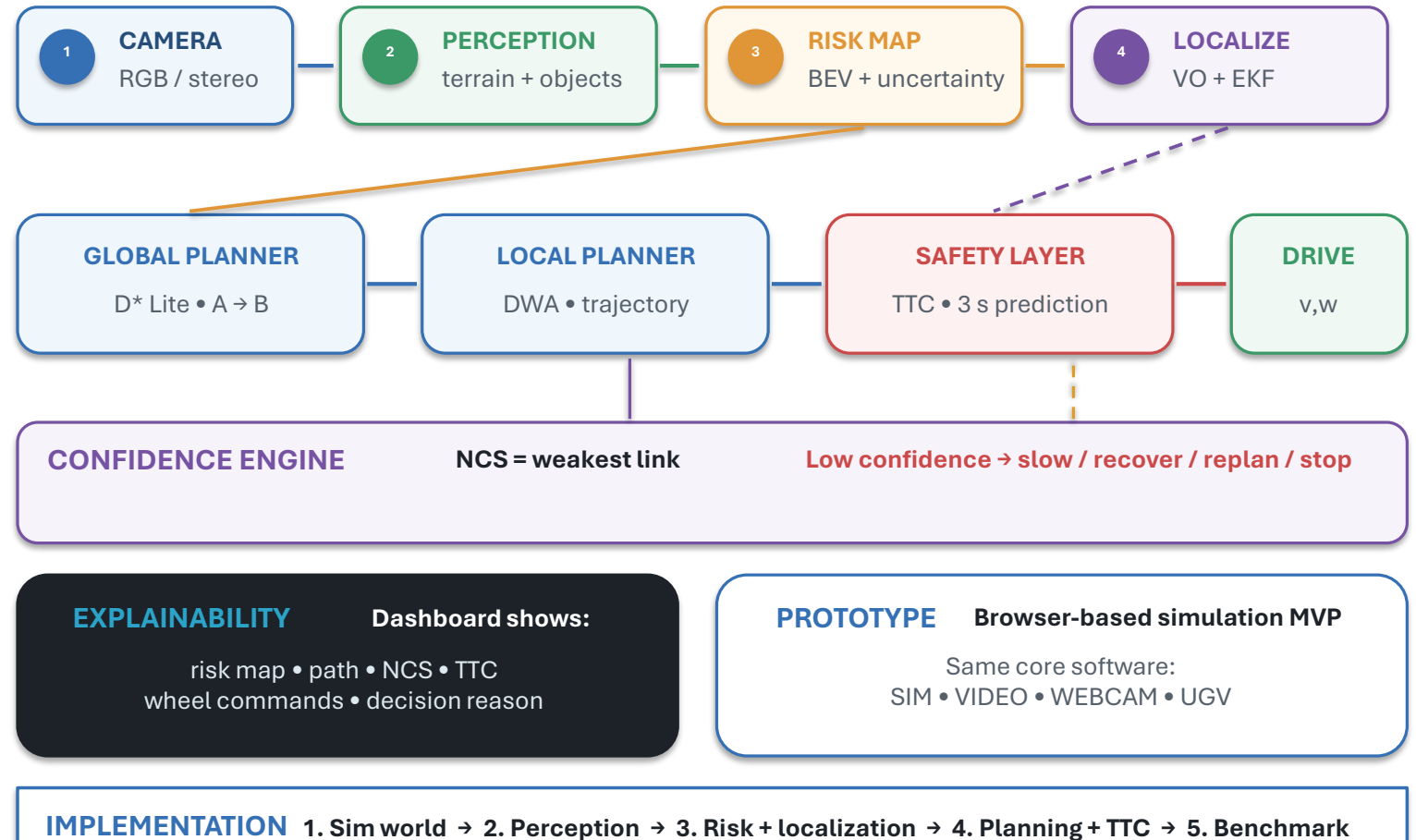
TECHNOLOGIES

- 1 AI / COMPUTER VISION**
Python • PyTorch • OpenCV • ONNX Runtime
- 2 PERCEPTION**
SegFormer-B0 + YOLOv8n / YOLO11n
- 3 LOCALIZATION**
Stereo VO + EKF • RTAB-Map / ORB-SLAM3
- 4 PLANNING & SAFETY**
D* Lite • DWA • TTC safety layer
- 5 ROBOTICS**
ROS 2 Jazzy • Gazebo Harmonic
- 6 DASHBOARD**
FastAPI + WebSocket • React + TypeScript

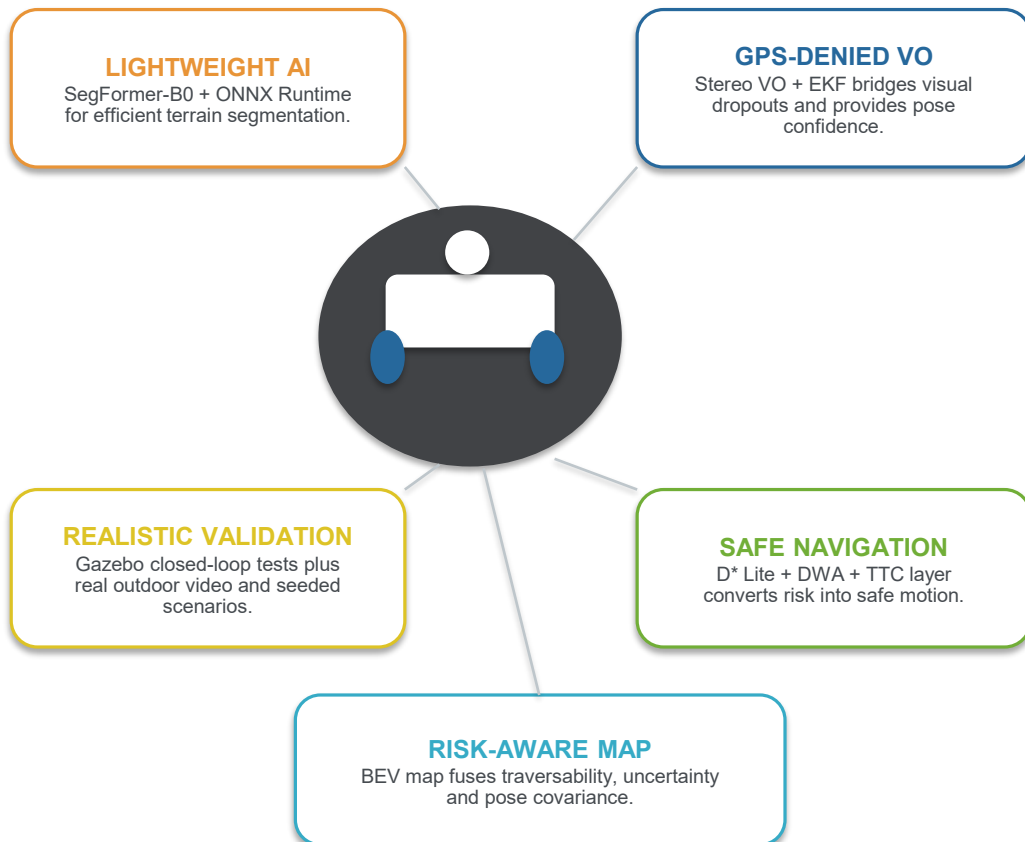
IMPLEMENTATION METHODOLOGY

Simulate → perceive → map → localize → plan → verify → benchmark

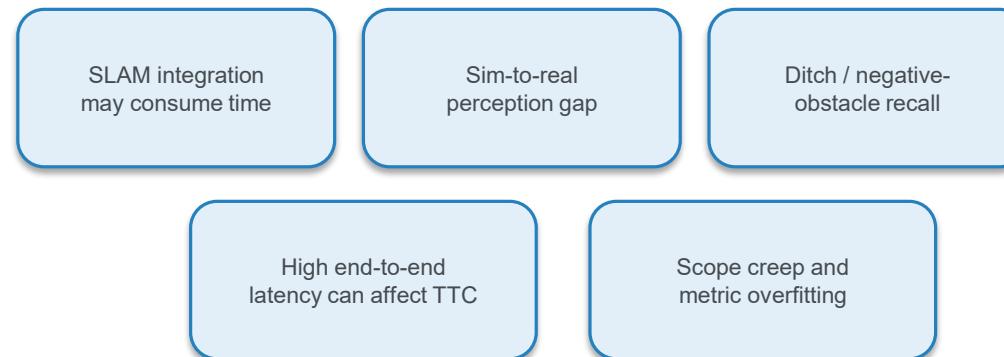
PROCESS FLOW ARCHITECTURE



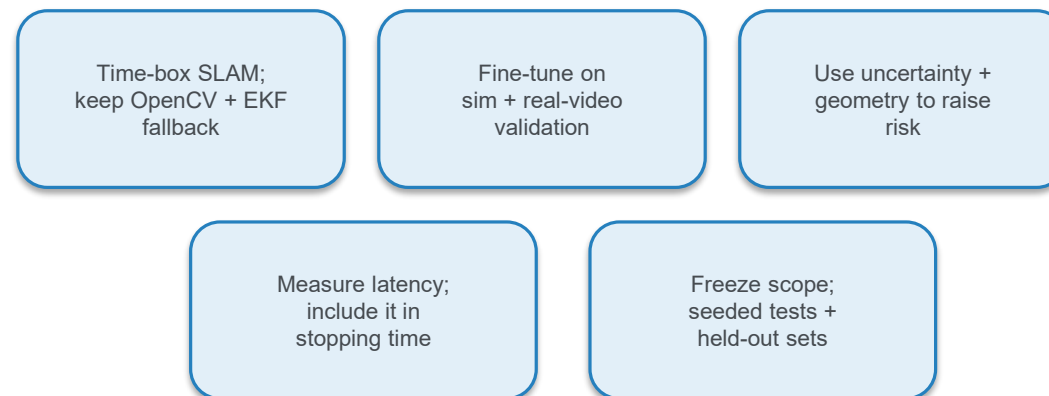
Feasibility of VISTRA



Potential Challenges & Risks



Strategies to Overcome Challenges



Evidence: baseline comparison • seeded scenarios • ablations • FPS/latency • recovery metrics

Benefits of VISTRA

Safer Navigation

Risk-aware planning and TTC safety reduce collision exposure.



Robust Autonomy

Confidence-gated behavior slows, recovers and safe-stops when uncertain.



Practical Validation

Seeded simulation, real outdoor video and baseline comparison support evidence.

Explainable Decisions

Dashboard shows risk, confidence, state changes and replan events.

V

Target Users / Stakeholders



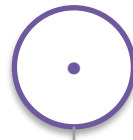
UGV
Operators



Autonomy
Developers



Safety &
Validation



R&D /
Simulation



System
Integrators

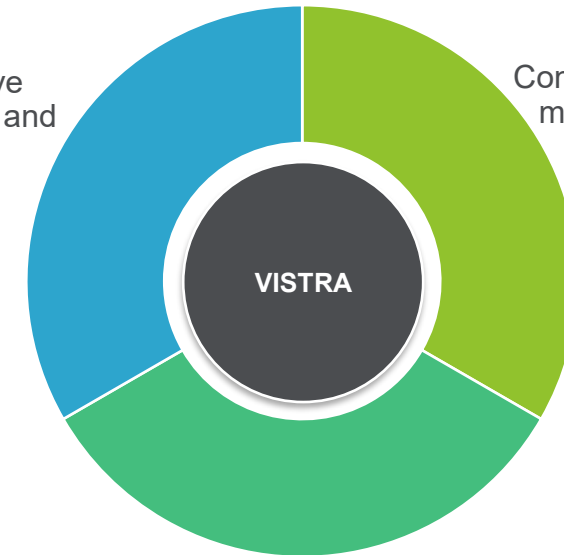
Benefits of the Solution

Safety

TTC-based predictive collision avoidance and safe-stop behavior.

Reliability

Confidence engine, recovery manager and pose-aware risk inflation.



Transparency

Live dashboard explains risk, confidence, replans and wheel commands.

- Details Links of the reference and research work

VERIFIED SOURCES USED BY VISTRA

1. SegFormer

Xie et al. (2021) | Semantic segmentation backbone

<https://arxiv.org/abs/2105.15203>

2. ORB-SLAM3

Campos et al. (2020) | Visual / visual-inertial SLAM

<https://arxiv.org/abs/2007.11898>

3. D* Lite

Koenig & Likhachev (2002) | Incremental global replanning

<https://ocs.aaai.org/Library/AAAI/2002/aaai02-072.php>

4. Dynamic Window Approach

Fox, Burgard & Thrun (1997) | Local collision avoidance

<https://publications.ri.cmu.edu/the-dynamic-window-approach-to-collision-avoidance>

5. RUGD Dataset

Wigness et al. (IROS 2019) | Unstructured outdoor driving data

https://rugd.vision/pdfs/RUGD_IROS2019.pdf

6. RELLIS-3D

Jiang et al. | Multi-modal off-road robotics dataset

<https://github.com/unmannedlab/RELLIS-3D>

7. ROS 2 Jazzy

Open Robotics | Robotics middleware and integration

<https://docs.ros.org/en/jazzy/>

8. Gazebo Harmonic

Open Robotics | Closed-loop UGV simulation

<https://gazebo-sim.org/docs/harmonic/getstarted/>

9. ONNX Runtime

Microsoft | Model inference/deployment runtime

<https://onnxruntime.ai/docs/>

Additional
Reference Link



MVP link: <https://vistara-blue.vercel.app/>

github Repo link : https://github.com/Zerovion/SIH_26126